

P8131 Homework 6

Yongyan Liu (yl6107)

2026-04-06

Problem 1

Suppose we have the following statistical model: $Y_{ij} = \mu + b_i + e_{ij}$, where $i = 1, \dots, m$ and $j = 1, \dots, n$, $b_i \sim N(0, \sigma_b^2)$, $e_{ij} \sim N(0, \sigma_e^2)$, b_i and e_{ij} are statistically independent for each i and j , and e_{ij} and e_{ik} are statistically independent for any two values $j, k = 1, \dots, n$ and $j \neq k$. Find the variance of Y_{ij} and the covariance and correlation between any two values Y_{ij} and Y_{ik} , $j \neq k$. Which one of the popular covariance patterns does this correspond to?

Variance of Y_{ij}

Since μ is a constant, it contributes zero variance. By the independence of b_i and e_{ij} :

$$\text{Var}(Y_{ij}) = \text{Var}(\mu + b_i + e_{ij}) = \text{Var}(b_i + e_{ij}) = \text{Var}(b_i) + \text{Var}(e_{ij}) = \sigma_b^2 + \sigma_e^2.$$

Covariance between Y_{ij} and Y_{ik} , $j \neq k$

$$\begin{aligned}\text{Cov}(Y_{ij}, Y_{ik}) &= \text{Cov}(b_i + e_{ij}, b_i + e_{ik}). \\ &= \text{Cov}(b_i, b_i) + \text{Cov}(b_i, e_{ik}) + \text{Cov}(e_{ij}, b_i) + \text{Cov}(e_{ij}, e_{ik}).\end{aligned}$$

By mutual independence: $\text{Cov}(b_i, e_{ik}) = 0$, $\text{Cov}(e_{ij}, b_i) = 0$, and $\text{Cov}(e_{ij}, e_{ik}) = 0$ (since $j \neq k$). Therefore:

$$\text{Cov}(Y_{ij}, Y_{ik}) = \text{Var}(b_i) = \sigma_b^2.$$

Correlation between Y_{ij} and Y_{ik} , $j \neq k$

Since $\text{Var}(Y_{ij}) = \text{Var}(Y_{ik}) = \sigma_b^2 + \sigma_e^2$ for all j :

$$\text{Corr}(Y_{ij}, Y_{ik}) = \frac{\text{Cov}(Y_{ij}, Y_{ik})}{\sqrt{\text{Var}(Y_{ij}) \text{Var}(Y_{ik})}} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_e^2}.$$

Covariance pattern

The correlation $\rho = \sigma_b^2 / (\sigma_b^2 + \sigma_e^2)$ is constant for all pairs $j \neq k$. The covariance matrix has the form:

$$\text{Cov}(\mathbf{Y}_i) = (\sigma_b^2 + \sigma_e^2) \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{pmatrix}.$$

This is the **compound symmetry** (also called exchangeable) covariance pattern.

Problem 2

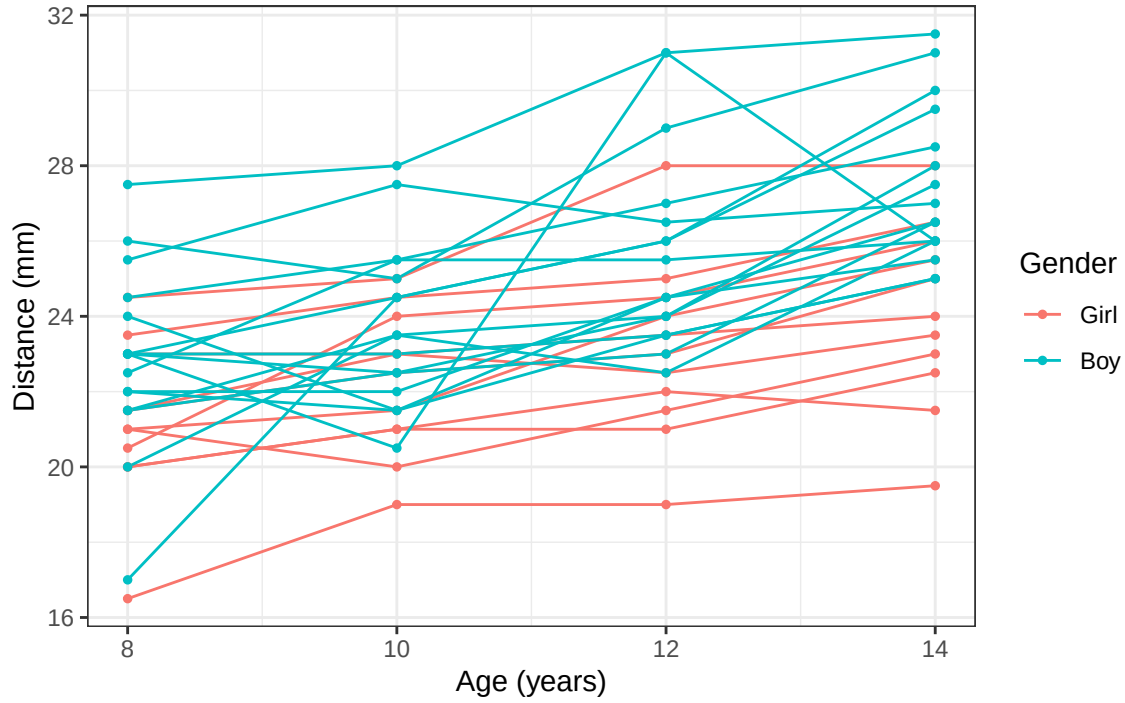
A study was conducted involving 27 children, 16 boys and 11 girls. On each child, the distance (mm) from the center of the pituitary to the pterygomaxillary fissure was made at ages 8, 10, 12, and 14. The goal was to study how distance is affected by age and gender.

```
dental <- read.table("HW6-dental.txt", header = TRUE)
dental$Gender <- factor(dental$Gender, levels = c(0, 1),
                        labels = c("Girl", "Boy"))
```

(a) Spaghetti plot

```
ggplot(dental, aes(x = Age, y = Distance,
                   group = Child, color = Gender)) +
  geom_line() +
  geom_point(size = 1) +
  labs(x = "Age (years)", y = "Distance (mm)",
       title = "Spaghetti Plot of Dental Distance by Age and Gender",
       color = "Gender") +
  theme_bw()
```

Spaghetti Plot of Dental Distance by Age and Gender



Distance generally increases with age for both boys and girls. Boys tend to have slightly larger distances and somewhat steeper growth trajectories than girls. There is considerable between-subject variability within each gender group.

(b) Marginal form of the mixed model

Assume the model:

$$Y_{ij} = \beta_0 + a_i + b_0 \cdot I(\text{sex}_i = 0) + b_1 \cdot I(\text{sex}_i = 1) + \beta_1 \cdot \text{age}_{ij} + e_{ij},$$

where $a_i \sim N(0, \sigma_a^2)$, $b_k \sim N(0, \sigma_b^2)$ ($k = 0, 1$) are random coefficients, and $e_{ij} \sim N(0, \sigma_e^2)$; all mutually independent.

Marginal mean

Taking the expectation of both sides:

$$E(Y_{ij}) = \beta_0 + E(a_i) + E(b_0) \cdot I(\text{sex}_i = 0) + E(b_1) \cdot I(\text{sex}_i = 1) + \beta_1 \cdot \text{age}_{ij} + E(e_{ij}).$$

Since a_i , b_0 , b_1 , and e_{ij} are all zero-mean random variables:

$$E(Y_{ij}) = \beta_0 + \beta_1 \cdot \text{age}_{ij}.$$

Marginal variance

For a given subject i , exactly one of $I(\text{sex}_i = 0)$ or $I(\text{sex}_i = 1)$ equals 1. The random components contributing to Y_{ij} are a_i , one of b_0 or b_1 (depending on sex), and e_{ij} . By mutual independence:

- If $\text{sex}_i = 0$: $\text{Var}(Y_{ij}) = \text{Var}(a_i) + \text{Var}(b_0) + \text{Var}(e_{ij}) = \sigma_a^2 + \sigma_b^2 + \sigma_e^2$.
- If $\text{sex}_i = 1$: $\text{Var}(Y_{ij}) = \text{Var}(a_i) + \text{Var}(b_1) + \text{Var}(e_{ij}) = \sigma_a^2 + \sigma_b^2 + \sigma_e^2$.

Since b_0 and b_1 share the same variance σ_b^2 :

$$\text{Var}(Y_{ij}) = \sigma_a^2 + \sigma_b^2 + \sigma_e^2$$

for all subjects regardless of gender.

Marginal covariance

For two observations $j \neq k$ on the same subject i , stripping the constant terms (which do not contribute to covariance) and noting that for a given subject exactly one of b_0 or b_1 applies (denote it b_{sex_i}):

$$\text{Cov}(Y_{ij}, Y_{ik}) = \text{Cov}(a_i + b_{\text{sex}_i} + e_{ij}, a_i + b_{\text{sex}_i} + e_{ik}).$$

By mutual independence of a_i , b_{sex_i} , e_{ij} , and e_{ik} : all cross-terms vanish. Therefore:

$$\text{Cov}(Y_{ij}, Y_{ik}) = \text{Var}(a_i) + \text{Var}(b_{\text{sex}_i}) = \sigma_a^2 + \sigma_b^2, \quad j \neq k.$$

The marginal covariance matrix is therefore:

$$\text{Cov}(\mathbf{Y}_i) = \begin{pmatrix} \sigma_a^2 + \sigma_b^2 + \sigma_e^2 & \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 \\ \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 + \sigma_e^2 & \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 \\ \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 + \sigma_e^2 & \sigma_a^2 + \sigma_b^2 \\ \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 + \sigma_e^2 \end{pmatrix},$$

which is a compound symmetry structure with correlation $\rho = (\sigma_a^2 + \sigma_b^2) / (\sigma_a^2 + \sigma_b^2 + \sigma_e^2)$.

(c) Fit marginal models with three covariance structures

Marginal model: $E(Y_{ij}) = \beta_0 + \beta_1 \cdot \text{sex}_i + \beta_2 \cdot \text{age}_{ij}$, $\text{Var}(\mathbf{Y}_i) = \Sigma$.

```
# Recode Gender as numeric for the marginal model
dental$sex <- ifelse(dental$Gender == "Boy", 1, 0)
```

(i) Compound symmetry

```
fit_cs <- gls(Distance ~ sex + Age,
              correlation = corCompSymm(form = ~ 1 | Child),
              data = dental, method = "REML")
summary(fit_cs)
```

```
## Generalized least squares fit by REML
##   Model: Distance ~ sex + Age
##   Data: dental
##           AIC       BIC    logLik
```

```
##    447.5125 460.7823 -218.7563
##
## Correlation Structure: Compound symmetry
## Formula: ~1 | Child
## Parameter estimate(s):
##      Rho
## 0.6144914
##
## Coefficients:
##              Value Std.Error   t-value p-value
## (Intercept) 15.385690 0.8959848 17.171820 0.0000
## sex          2.321023 0.7614169 3.048294 0.0029
## Age          0.660185 0.0616059 10.716263 0.0000
##
## Correlation:
##      (Intr) sex
## sex -0.504
## Age -0.756 0.000
##
## Standardized residuals:
##      Min      Q1      Med      Q3      Max
## -2.59712955 -0.64544226 -0.02540005 0.51680604 2.32947531
##
## Residual standard error: 2.305697
## Degrees of freedom: 108 total; 105 residual
```

(ii) Exponential

```
fit_exp <- gls(Distance ~ sex + Age,
               correlation = corExp(form = ~ 1 | Child),
               data = dental, method = "REML")
summary(fit_exp)
```

```
## Generalized least squares fit by REML
## Model: Distance ~ sex + Age
## Data: dental
##      AIC      BIC    logLik
## 455.4483 468.7181 -222.7241
##
## Correlation Structure: Exponential spatial correlation
## Formula: ~1 | Child
## Parameter estimate(s):
##      range
## 2.133938
##
## Coefficients:
##              Value Std.Error   t-value p-value
## (Intercept) 15.459995 1.1309319 13.670138 0e+00
## sex          2.418714 0.6933441 3.488476 7e-04
## Age          0.652960 0.0906420 7.203723 0e+00
##
## Correlation:
```

```
##      (Intr) sex
## sex -0.363
## Age -0.882  0.000
##
## Standardized residuals:
##      Min      Q1      Med      Q3      Max
## -2.65148775 -0.69592567 -0.06214639  0.48659340  2.29666951
##
## Residual standard error: 2.301495
## Degrees of freedom: 108 total; 105 residual
```

(iii) Autoregressive AR(1)

```
fit_ar1 <- gls(Distance ~ sex + Age,
               correlation = corAR1(form = ~ 1 | Child),
               data = dental, method = "REML")
summary(fit_ar1)
```

```
## Generalized least squares fit by REML
## Model: Distance ~ sex + Age
## Data: dental
##      AIC      BIC    logLik
## 455.4483 468.7181 -222.7241
##
## Correlation Structure: AR(1)
## Formula: ~1 | Child
## Parameter estimate(s):
##      Phi
## 0.6258671
##
## Coefficients:
##      Value Std.Error   t-value p-value
## (Intercept) 15.459995 1.1309319 13.670138 0e+00
## sex          2.418714 0.6933441  3.488476 7e-04
## Age          0.652960 0.0906420  7.203723 0e+00
##
## Correlation:
##      (Intr) sex
## sex -0.363
## Age -0.882  0.000
##
## Standardized residuals:
##      Min      Q1      Med      Q3      Max
## -2.65148770 -0.69592566 -0.06214639  0.48659339  2.29666947
##
## Residual standard error: 2.301495
## Degrees of freedom: 108 total; 105 residual
```

Comparison of coefficient estimates

```
coef_table <- data.frame(
  CS = round(coef(fit_cs), 4),
  Exp = round(coef(fit_exp), 4),
  AR1 = round(coef(fit_ar1), 4)
)
rownames(coef_table) <- c("Intercept (beta_0)", "Sex (beta_1)",
  "Age (beta_2)")
knitr::kable(coef_table, caption = "Fixed effect estimates across models")
```

Table 1: Fixed effect estimates across models

	CS	Exp	AR1
Intercept (beta_0)	15.3857	15.4600	15.4600
Sex (beta_1)	2.3210	2.4187	2.4187
Age (beta_2)	0.6602	0.6530	0.6530

The fixed effect estimates are very similar across all three covariance structures.

Comparison of covariance estimates

```
# Compound symmetry
cat("=== Compound Symmetry ===\n")

## === Compound Symmetry ===

cat("Residual sigma:", fit_cs$sigma, "\n")

## Residual sigma: 2.305697

cat("Rho:", coef(fit_cs$modelStruct$corStruct, unconstrained = FALSE),
  "\n")

## Rho: 0.6144914

cs_cor <- corMatrix(fit_cs$modelStruct$corStruct)[[1]]
cat("Correlation matrix:\n")

## Correlation matrix:

print(round(cs_cor, 4))

##      [,1] [,2] [,3] [,4]
## [1,] 1.0000 0.6145 0.6145 0.6145
## [2,] 0.6145 1.0000 0.6145 0.6145
## [3,] 0.6145 0.6145 1.0000 0.6145
## [4,] 0.6145 0.6145 0.6145 1.0000
```

```
cat("\n=== Exponential ===\n")
```

```
##  
## === Exponential ===
```

```
cat("Residual sigma:", fit_exp$sigma, "\n")
```

```
## Residual sigma: 2.301495
```

```
cat("Range:", coef(fit_exp$modelStruct$corStruct, unconstrained = FALSE),  
    "\n")
```

```
## Range: 2.133938
```

```
exp_cor <- corMatrix(fit_exp$modelStruct$corStruct)[[1]]  
cat("Correlation matrix:\n")
```

```
## Correlation matrix:
```

```
print(round(exp_cor, 4))
```

```
##      [,1] [,2] [,3] [,4]  
## [1,] 1.0000 0.6259 0.3917 0.2452  
## [2,] 0.6259 1.0000 0.6259 0.3917  
## [3,] 0.3917 0.6259 1.0000 0.6259  
## [4,] 0.2452 0.3917 0.6259 1.0000
```

```
cat("\n=== AR(1) ===\n")
```

```
##  
## === AR(1) ===
```

```
cat("Residual sigma:", fit_ar1$sigma, "\n")
```

```
## Residual sigma: 2.301495
```

```
cat("Phi:", coef(fit_ar1$modelStruct$corStruct, unconstrained = FALSE),  
    "\n")
```

```
## Phi: 0.6258671
```

```
ar1_cor <- corMatrix(fit_ar1$modelStruct$corStruct)[[1]]  
cat("Correlation matrix:\n")
```

```
## Correlation matrix:
```



```
print(round(ar1_cor, 4))
```

```
##           [,1]    [,2]    [,3]    [,4]
## [1,] 1.0000 0.6259 0.3917 0.2452
## [2,] 0.6259 1.0000 0.6259 0.3917
## [3,] 0.3917 0.6259 1.0000 0.6259
## [4,] 0.2452 0.3917 0.6259 1.0000
```

Summary of covariance parameters:

For this dataset, the time points are equally spaced (ages 8, 10, 12, 14 with spacing 2), so the AR(1) and exponential models can produce identical correlation structures. In contrast, compound symmetry assumes the same correlation regardless of time separation, which may be overly restrictive if correlations decay with increasing time lag.